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```
clc;
clear;
close all;
```

STRUCTURAL PARAMETERS (Table 1)

```
m1 = 10;    m2 = 10;
k1 = 10000; k2 = 10000;    % 10 kN/m → 10000 N/m
c1 = 31.76; c2 = 31.76;
```

```
M = [m1 0;
      0  m2];
```

```
K = [k1+k2  -k2;
      -k2    k2];
```

```
C = [c1+c2  -c2;
      -c2    c2];
```

```
influence = [1;1];
```

CONTINUOUS-TIME STATE-SPACE MODEL

State: $X = [y_1 \ y_2 \ \dot{y}_1 \ \dot{y}_2]'$

```
Ac = [ zeros(2)  eye(2);
      -M\K       -M\C ];
```

```
Bc = [ zeros(2,1);
      -influence ];
```

LOAD DATA

```
% Ground motion (50 Hz)
gm = load('HW1_Q2_El_Centro.txt');
```

```

t_gm = gm(:,1);
ug_ddot_raw = gm(:,2);

% Measurements (200 Hz)
meas = load('HW1_Q2_Measurements.txt');
t = meas(:,1);
y_meas = meas(:,2:3);      % displacement
a_meas = meas(:,4:5);      % absolute acceleration

dt = t(2) - t(1);
N = length(t);

% Interpolate ground motion to 200 Hz
ug_ddot = interp1(t_gm, ug_ddot_raw, t, 'linear','extrap');

```

DISCRETIZE SYSTEM (Exact Discretization)

```

Phi = expm(Ac*dt);
Gamma = (Phi - eye(4))*(Ac\Bc);

```

MEASUREMENT MODEL (ACCELERATION ONLY)

Absolute acceleration: $y_ddot + ug_ddot = -M^{-1} K y - M^{-1} C y_dot$

```

H = [-M\K    -M\C];

```

```

% Noise variances (Table 2)
var_a = [0.105  0.162];
R = diag(var_a);

```

```

Z_all = a_meas';

```

INITIAL CONDITIONS

```

X0 = zeros(4,1);      % structure initially at rest
P0 = eye(4)*1e-4;

```

RUN KALMAN FILTER – CASE 1: $Q = 0$

```

Q_zero = zeros(4);
X = X0;
P = P0;
X_est_zero = zeros(4,N);

for k = 1:N

    % Prediction
    X_pred = Phi*X + Gamma*ug_ddot(k);
    P_pred = Phi*P*Phi' + Q_zero;

```

```

% Update
Z = Z_all(:,k);
K_gain = P_pred*H'/(H*P_pred*H' + R);

X = X_pred + K_gain*(Z - H*X_pred);
P = (eye(4) - K_gain*H)*P_pred;

X_est_zero(:,k) = X;
end

```

RUN KALMAN FILTER – CASE 2: $Q \neq 0$

```

Q_small = eye(4)*1e-7; % small process noise
X = X0;
P = P0;
X_est_smallQ = zeros(4,N);

for k = 1:N

    % Prediction
    X_pred = Phi*X + Gamma*ug_ddot(k);
    P_pred = Phi*P*Phi' + Q_small;

    % Update
    Z = Z_all(:,k);
    K_gain = P_pred*H'/(H*P_pred*H' + R);

    X = X_pred + K_gain*(Z - H*X_pred);
    P = (eye(4) - K_gain*H)*P_pred;

    X_est_smallQ(:,k) = X;
end

```

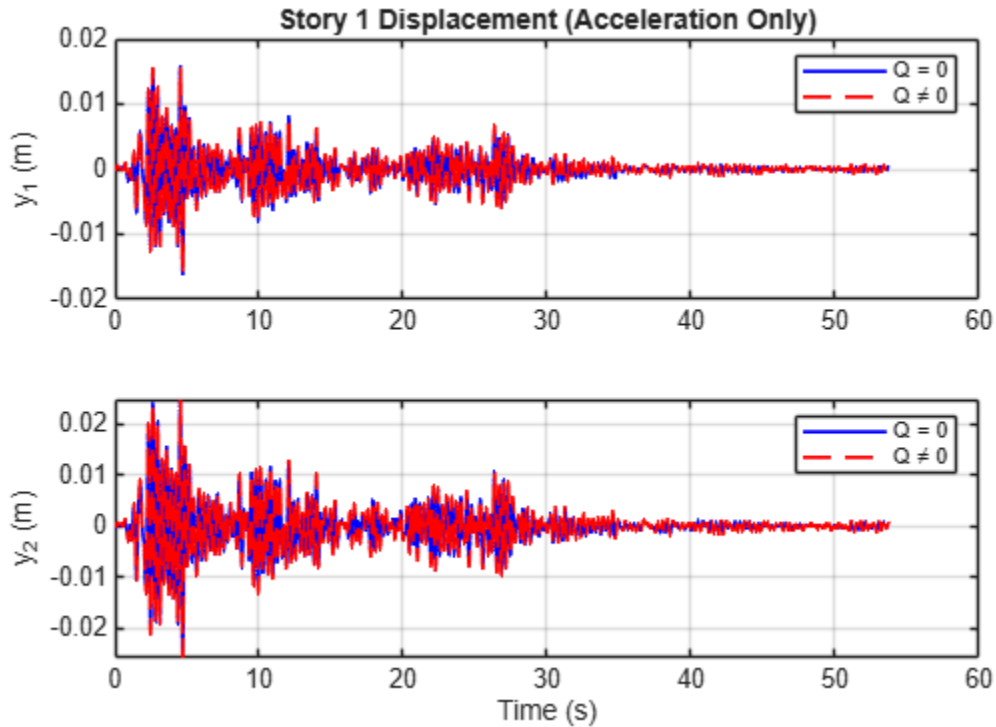
PLOT RESULTS – DISPLACEMENTS

```

figure;
subplot(2,1,1)
plot(t, X_est_zero(1,:), 'b', 'LineWidth',1.2); hold on;
plot(t, X_est_smallQ(1,:), 'r--', 'LineWidth',1.2);
title('Story 1 Displacement (Acceleration Only)');
ylabel('y_1 (m)');
legend('Q = 0', 'Q ≠ 0');
grid on

subplot(2,1,2)
plot(t, X_est_zero(2,:), 'b', 'LineWidth',1.2); hold on;
plot(t, X_est_smallQ(2,:), 'r--', 'LineWidth',1.2);
ylabel('y_2 (m)');
xlabel('Time (s)');
legend('Q = 0', 'Q ≠ 0');
grid on

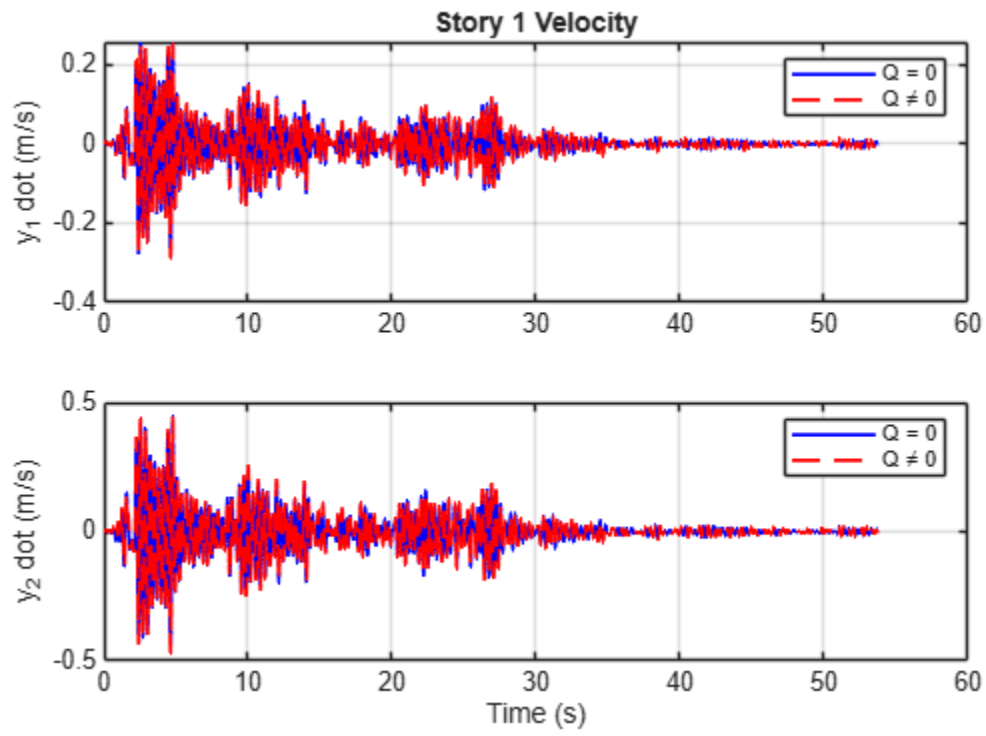
```



VELOCITY RESPONSE

```
figure;
subplot(2,1,1)
plot(t, X_est_zero(3,:), 'b', 'LineWidth',1.2); hold on;
plot(t, X_est_smallQ(3,:), 'r--', 'LineWidth',1.2);
title('Story 1 Velocity');
ylabel('y_1 dot (m/s)');
legend('Q = 0', 'Q ≠ 0');
grid on

subplot(2,1,2)
plot(t, X_est_zero(4,:), 'b', 'LineWidth',1.2); hold on;
plot(t, X_est_smallQ(4,:), 'r--', 'LineWidth',1.2);
ylabel('y_2 dot (m/s)');
xlabel('Time (s)');
legend('Q = 0', 'Q ≠ 0');
grid on
```



Published with MATLAB® R2025b